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## **DAY-1(SEPT 1)Class Work**

Part A — User and System Investigation (10 minutes)

Determine the username of the account you are currently using.

Display your UID, primary GID, and supplementary groups.

Display all users currently logged in to the system.

Determine what activities the currently logged-in users are performing.

Examine /etc/passwd and identify the following information for your current account:

Username

UID

GID

Home directory

Login shell

Determine how many user-account entries are present in /etc/passwd.

Display only the username and UID fields from /etc/passwd.

Find all accounts whose UID is 0. Explain the security significance of UID 0.

Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

```

anonymous@anonymous-supreme in ~ via v3.14.7
# } bash

anonymous@anonymous-supreme in ~ via v3.14.7
# } whoami
anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -un
anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
# } id
uid=1000(anonymous) gid=1000(anonymous) groups=1000(anonymous),3(sys),96(scanner),188(vboxusers),289(cups),944(docker),953(sambashare),970(realtime),985(video),991(lp),994(input),998(wheel)

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -u
1000

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -g
1000

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -G
1000 3 96 108 289 944 953 970 985 991 994 998

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -Gn
anonymous sys scanner vboxusers cups docker sambashare realtime video lp input wheel

anonymous@anonymous-supreme in ~ via v3.14.7
# } who
anonymous pts/0      2026-09-01 09:19 (:0)
anonymous pts/1      2026-09-01 09:54 (:0)

anonymous@anonymous-supreme in ~ via v3.14.7
# } w
09:56:50 up 38 min, 1 user, load average: 1.02, 1.09, 0.77
USER      TTY      LOGIN@   IDLE   JCPU   PCPU   WHAT
anonymou tty2      09:19    37:37   0.02s   0.02s /usr/bin/startplasma-wayland

anonymous@anonymous-supreme in ~ via v3.14.7
# } grep "^$(whoami):" /etc/passwd
anonymous:x:1000:1000:Anonymous:/home/anonymous:/bin/bash

anonymous@anonymous-supreme in ~ via v3.14.7
# } _

```

```

anonymous@anonymous-supreme in ~ via v3.14.7
# } grep "^$(whoami):" /etc/passwd
anonymous:x:1000:1000:Anonymous:/home/anonymous:/bin/bash

anonymous@anonymous-supreme in ~ via v3.14.7
# } [200~cut -d: -f1 /etc/passwd | grep "^$(whoami)$"~
find-the-command: "[200~cut" is not found locally, searching in repositories...
find-the-command: /var/cache/pkgfile/*.files are out of date, update? [Y/n] n
find-the-command: command not found: "[200~cut"

anonymous@anonymous-supreme in ~ via v3.14.7 took 2s79ms
[[Q]![*]] => *ERROR # cut -d: -f1 /etc/passwd | grep "^$(whoami)$"
anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -u
1000

anonymous@anonymous-supreme in ~ via v3.14.7
# } id -g
1000

anonymous@anonymous-supreme in ~ via v3.14.7
# } getent passwd "$(whoami)" | cut -d: -f6
/home/anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
# } getent passwd "$(whoami)" | cut -d: -f7
/bin/bash

anonymous@anonymous-supreme in ~ via v3.14.7
# } _

```

```
anonymous@anonymous-supreme in ~ via v3.14.7
/ ) awk -F: '{print $1, $3}' /etc/passwd
root 0
bin 1
daemon 2
mail 8
ftp 14
http 33
nobody 65534
dbus 81
systemd-coredump 980
systemd-network 979
systemd-oom 978
systemd-journal-remote 977
systemd-resolve 976
systemd-timesync 975
uuid 68
alpm 974
avahi 973
named 40
_talkd 972
rpc 32
rpcuser 34
tss 971
dnsmasq 967
fwupd 966
git 965
nm-openconnect 964
nm-openvpn 963
ldap 439
openvpn 962
passim 961
polkitd 102
rtkit 133
sddm 960
anonymous 1000
nvidia-persistenced 143
atftp 958
darkstat 957
radiusd 956
i2pd 955
sniffglue 954
stunnel 16
tinyproxy 186
tor 43
colord 952
cups 209
saned 951
pcscd 949
mosquitto 947
systemd-imds 946
plasma-login 945

anonymous@anonymous-supreme in ~ via v3.14.7
/ ) awk -F: '$3 == 0 {print $1, $3}' /etc/passwd
root 0

anonymous@anonymous-supreme in ~ via v3.14.7
/ ) cat /etc/passwd
```

```
File: /etc/passwd
root:x:0:0::/root:/usr/bin/bash
bin:x:1:1::/usr/bin/nologin
daemon:x:2:2::/usr/bin/nologin
mail:x:8:12::/var/spool/mail:/usr/bin/nologin
ftp:x:14:11::/srv/ftp:/usr/bin/nologin
http:x:33:33::/srv/http:/usr/bin/nologin
nobody:x:65534:65534:Kernel Overflow User:/usr/bin/nologin
dbus:x:81:81:System Message Bus:/usr/bin/nologin
systemd-coredump:x:980:980:systemd Core Dumper:/usr/bin/nologin
systemd-network:x:979:979:systemd Network Management:/usr/bin/nologin
systemd-oom:x:978:978:systemd Userspace OOM Killer:/usr/bin/nologin
systemd-journal-remote:x:977:977:systemd Journal Remote:/usr/bin/nologin
systemd-resolve:x:976:976:systemd Resolver:/usr/bin/nologin
systemd-timesync:x:975:975:systemd Time Synchronization:/usr/bin/nologin
uidd:x:68:68:uidd daemon:/usr/bin/nologin
alpm:x:974:974:Arch Linux Package Management:/usr/bin/nologin
avahi:x:973:973:Avahi mDNS/DNS-SD daemon:/usr/bin/nologin
:
```

```
anonymous@anonymous-supreme in ~ via 🐉 v3.14.7 took 1m9s510ms
➜ sudo cat /etc/shadow
[sudo] password for anonymous:
root:$y$j9T$y7/mltLUBKL3fam8tofVG/$7qW4tvyW00wonXTJT6sCmE2QlcYyB/iDjQca50eGLa
4:20174::::::
bin:!*:20155::::::
daemon:!*:20155::::::
```

## Part B — Creating and Verifying Users (20 Minutes)

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

Verify that all three accounts have been added to `/etc/passwd`.

Determine the UID assigned to each newly created user.

Determine the home directory and default shell assigned to alice.

Inspect `/home` and verify that the required home directories were created.

Set passwords for all three accounts.

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```

anonymous@anonymous-supreme in ~ via v3.14.7 as  took 11s25ms
/ ) sudo useradd -m alice

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) sudo useradd -m bob

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) sudo useradd -m charlie

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1001:1001::/home/alice:/usr/bin/bash
bob:x:1002:1002::/home/bob:/usr/bin/bash
charlie:x:1003:1003::/home/charlie:/usr/bin/bash

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice)

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie)

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) getent passwd alice
alice:x:1001:1001::/home/alice:/usr/bin/bash

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) ls -la /home
drwxr-xr-x - root      1 Sep 10:17 .
dr-xr-xr-x - root      23 Oct 2025 ..
drwx----- - alice     1 Sep 10:17 alice
drwx----- - anonymous 1 Sep 09:19 anonymous
drwx----- - bob       1 Sep 10:17 bob
drwx----- - charlie   1 Sep 10:17 charlie

anonymous@anonymous-supreme in ~ via v3.14.7 as 
/ ) sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

anonymous@anonymous-supreme in ~ via v3.14.7 as  took 26s88ms
/ ) sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully

anonymous@anonymous-supreme in ~ via v3.14.7 as  took 7s615ms
/ ) sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully

```

```

anonymous@anonymous-supreme in ~ via v3.14.7 as  took 12s499ms
/ ) su - alice
Password:
-bash: n: command not found
[alice@anonymous-supreme ~]$ whoami
alice
[alice@anonymous-supreme ~]$ id
uid=1001(alice) gid=1001(alice) groups=1001(alice)
[alice@anonymous-supreme ~]$ pwd
/home/alice
[alice@anonymous-supreme ~]$ exit
logout

anonymous@anonymous-supreme in ~ via v3.14.7 as  took 32s789ms
/ ) whoami
anonymous

```

## Part C — Group Management and Access Control (30 Minutes)

Create two groups:

security and developers

Add alice and charlie to the security group.

Add bob to the developers group.

Without switching accounts, verify the group membership of all three users.

Display the entries corresponding to security and developers from /etc/group.

Add bob to the security group without removing him from the developers group.

Verify that bob now belongs to both groups.

```
anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) sudo groupadd security

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) sudo groupadd developers

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) getent group security
security:x:1004:

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) getent group developers
developers:x:1005:

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) sudo usermod -aG security alice

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) sudo usermod -aG security charlie

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) sudo usermod -aG developers bob

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) id alice id bob id charlie
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(security)
id: 'id': no such user
uid=1002(bob) gid=1002(bob) groups=1002(bob),1005(developers)
id: 'id': no such user
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1004(security)

anonymous@anonymous-supreme in ~ via v3.14.7 as 
[*] # grep -E '^(security|developers):' /etc/group
security:x:1004:alice,charlie
developers:x:1005:bob

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) sudo usermod -aG security bob

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) groups bob
bob : bob security developers

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) grep -E '^(security|developers):' /etc/group
security:x:1004:alice,charlie,bob
developers:x:1005:bob

anonymous@anonymous-supreme in ~ via v3.14.7 as 
# ) _
```